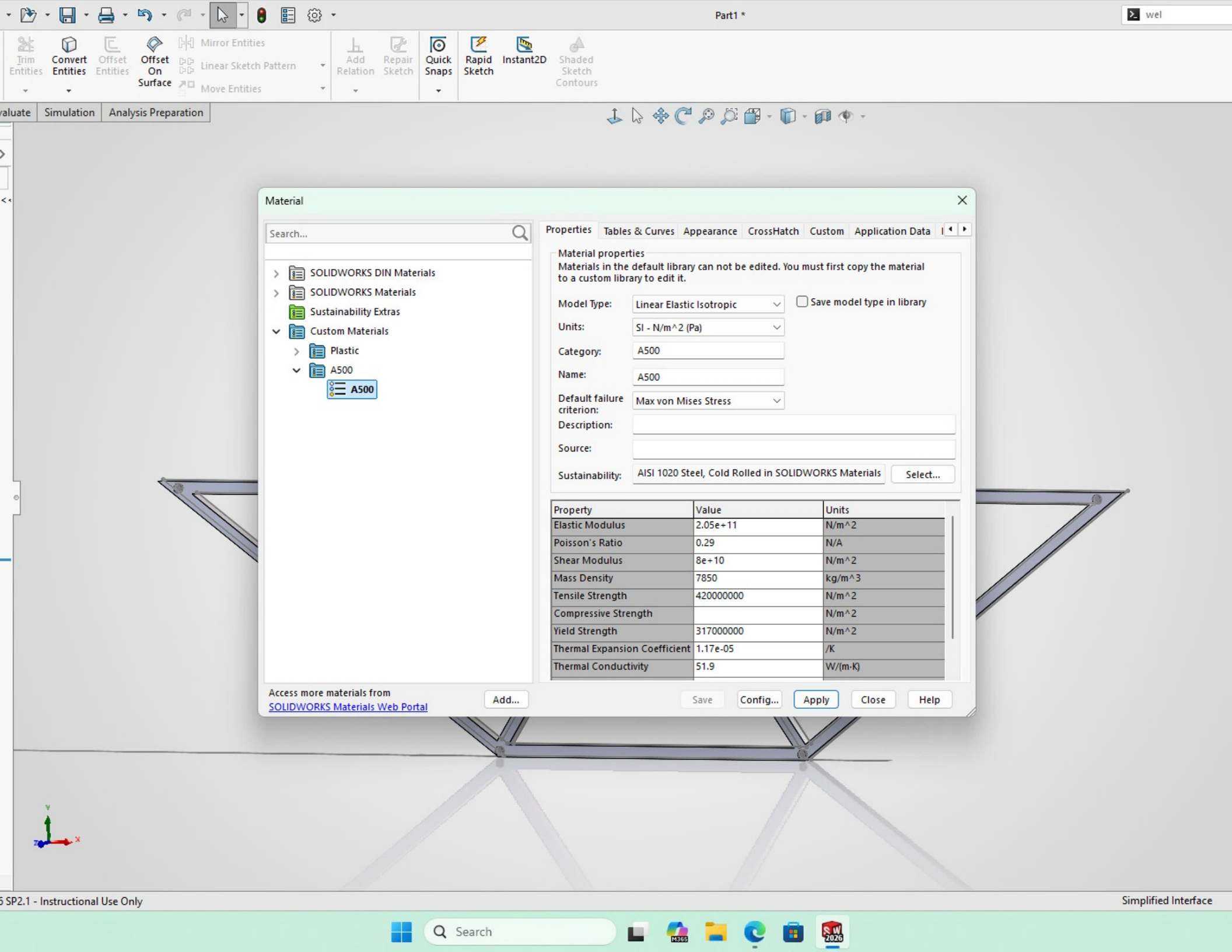




Material

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 - A500

Properties Tables & Curves Appearance CrossHatch Custom Application Data

Material properties
Materials in the default library can not be edited. You must first copy the material to a custom library to edit it.

Model Type: Linear Elastic Isotropic ☐ Save model type in library

Units: SI - N/m² (Pa)

Category: A500

Name: A500

Default failure criterion: Max von Mises Stress

Description:

Source:

Sustainability: AISI 1020 Steel, Cold Rolled in SOLIDWORKS Materials

Property	Value	Units
Elastic Modulus	2.05e+11	N/m ²
Poisson's Ratio	0.29	N/A
Shear Modulus	8e+10	N/m ²
Mass Density	7850	kg/m ³
Tensile Strength	420000000	N/m ²
Compressive Strength		N/m ²
Yield Strength	317000000	N/m ²
Thermal Expansion Coefficient	1.17e-05	/K
Thermal Conductivity	51.9	W/(m·K)

Access more materials from
[SOLIDWORKS Materials Web Portal](#)